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ROTARY ENGINE LUBRICATION SYSTEM

Abstract

A rotary piston internal combustion engine includes an engine housing, an output shaft configured to rotate relative to the engine housing about an axis for transmitting power to parts to be operated, a rotary piston mounted eccentrically on the output shaft, a first bearing configured to support and facilitate rotation of the output shaft, the first bearing being sandwiched between a first pair of seals and a second bearing configured to support and facilitate rotation of the rotary piston, the second bearing being sandwiched between a second pair of seals, wherein the output shaft defines an internal path which has a first section to allow a lubricating substance to flow between the first bearing and the second bearing, in use.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to rotary piston internal combustion engines. More particularly, the invention relates to rotary piston internal combustion engines having a configuration which provides for improved lubrication efficiency of internal bearings and sliding surfaces.

BACKGROUND

[0002] It is known that rotary piston internal combustion engines, such as the Wankel engine, include an oval-like epitrochoidal cavity formed within the engine housing, a rotary piston and an output shaft. The rotary piston has a substantially triangular shape with convex arcuate flanks. Apex seals are located at the three apices of the rotary piston. During operation, the apex seals maintain physical contact with an inner peripheral surface of the cavity, thereby forming three combustion chambers. The cavity is provided with inlet and exhaust ports, the inlet port supplying a fuel and/or air charge to the cavity and the exhaust port venting exhaust gases to atmosphere after combustion has occurred in the combustion chambers. Combustion of the fuel causes expansion of combustion gases resulting in increased pressure in the combustion chambers, thereby resulting in rotation of the rotary piston relative to the engine housing. For every rotation of the rotary piston there are three rotations of the output shaft. A lubricating substance, such as oil, can be introduced to the engine to prevent overheating and seizing by lubricating the moving parts (e.g. bearings and seals), thus extending the lifetime of the engine.

[0003] One known method of lubricating moving parts of a rotary engine can be found in EP2240671, which discloses a metering pump that supplies oil to a cooling circuit.

[0004] The oil is carried by the circulating cooling or combustion gas to lubricate various components, such as bearings, gears, and the internal sliding surfaces of the rotary piston. A proportion of oil then passes out of the rotary piston, through external ducting, through a cooling heat exchanger and pump to be returned to the rotary piston. Circulating oil can also enter the combustion chambers of the engine as part of an air induction charge. Consequently, all oil pumped into this type of engine is ultimately burnt and exhausted to the atmosphere.

[0005] A disadvantage of the lubricating system described in EP2240671 is that a large amount of oil introduced to the engine is consumed. For example, the 225CS™ (which is a 40 BHP rotary engine manufactured by Advanced Innovative Engineering of Lichfield in the UK which has a lubricating system taught by EP2240671) consumes oil at a rate of approximately 150 cc/hr. In this engine oil distribution is indiscriminate and dependent on engine dynamics and system pressures. Consequently, oil reaches components that do not require lubrication. Therefore, a large volume of lubricating substance is needed to ensure all the moving components of the engine are sufficiently lubricated. Furthermore, the oil can become entrained with impurities, e.g., carbon build up from the sliding surfaces of the rotary piston, which means that impure oil can be transferred to other component parts such as the bearings due to the fact that some of the oil can recirculate.

[0006] Vehicles or devices comprising such engines can have high hydrocarbon emissions due to the amount of oil that is burnt. This means that owners of such vehicles may be subjected to charges, e.g. clean air charges, in countries that have strict policies on vehicle emissions. Due to

global warming, more stringent restrictions may be adopted by many countries, which may lead to vehicles having high carbon emissions being prohibited in some countries. Moreover, maintenance costs are not insignificant since a large amount of oil is required to keep these engines healthy. Furthermore, a large capacity oil reservoir may be needed to store the required oil which is not ideal.

[0007] It is a non-exclusive object of the present invention to overcome or at least substantially alleviate one or more of the problems associated with the prior art and/or mentioned above.

[0008] One objective is to maintain lubrication performance whilst reducing oil consumption.

SUMMARY OF INVENTION

[0009] According to a first aspect of the invention, there is provided a rotary piston internal combustion engine including: [0010] an engine housing; [0011] an output shaft configured to rotate relative to the engine housing about an axis for transmitting power to parts to be operated; [0012] a rotary piston mounted eccentrically on the output shaft; [0013] a first bearing configured to support and facilitate rotation of the output shaft, the first bearing being sandwiched between a first pair of seals; and [0014] a second bearing configured to support and facilitate rotation of the rotary piston, the second bearing being sandwiched between a second pair of seals; [0015] wherein the output shaft defines an internal path comprising a first section to allow a lubricating substance (e.g. oil) to flow between the first bearing and the second bearing, in use.

[0016] The rotary piston internal combustion engine may be configured so that the lubricating substance flows from the first bearing to the second bearing.

[0017] Optionally, the internal path comprises a second section to allow the lubricating substance to flow between the second bearing and a third bearing, in use.

[0018] The rotary piston internal combustion engine may be configured so that the lubricating substance flows from the second bearing to the third bearing.

[0019] Optionally, the output shaft defines a bore which at least partially extends along the axis.

The rotary piston internal combustion engine may further comprise a first barrier member. The first barrier member may be located in the bore between the first section and the second section. The first barrier member may be configured to direct the flow of the lubricating substance from the first section to the second bearing.

[0020] Optionally, the first barrier member is configured to prevent the flow of the lubricating substance along the bore and past the first barrier member.

[0021] Optionally, the rotary piston internal combustion engine further comprises a second barrier member. The second barrier member may be located in the bore and be configured to direct the flow of the lubricating substance from the second section to the third bearing.

[0022] Optionally, the second barrier member is configured to prevent the flow of the lubricating substance along the bore and past the second barrier member.

[0023] Optionally, the rotary piston internal combustion engine further comprises a rod located within the bore.

[0024] Optionally, the rod comprises one or both of the first barrier member and the second barrier member.

[0025] Optionally, the rod is biased towards the first bearing. The rod may be biased by a resilient member, such as a spring, e.g. a coil spring. The resilient member may be located within the bore.

[0026] Optionally, the third bearing is configured to support and facilitate rotation of the output shaft.

[0027] Optionally, the third bearing is in fluid communication with a thrust bearing.

[0028] Optionally, the third bearing and the thrust bearing are configured so that, in use, the lubricating substance is received by the thrust bearing from the third bearing along a direction that is parallel to the axis.

[0029] Optionally, the thrust bearing comprises a first side and a second side, the second side being spaced apart from the first side along the axis, wherein the first side includes an inlet for receiving

the lubricating substance from the third bearing and the second side includes an outlet for the lubricating substance to exit the thrust bearing.

[0030] Optionally, the third bearing is sandwiched between the thrust bearing and a third seal.

[0031] Optionally, the first bearing and/or the third bearing include(s): [0032] a ring; [0033] a retainer located within the ring and mounted coaxially relative thereto; and [0034] a rolling element, [0035] wherein the rolling element is held relative to the ring by the retainer.

[0036] Optionally, the retainer defines an aperture which extends from an outer surface to an inner surface thereof, wherein the aperture receives at least a part of the rolling element.

[0037] Optionally, the ring includes an inlet for receiving the lubricating substance.

[0038] Optionally, the ring includes a channel which extends around an outer surface thereof, the channel being in fluid communication with the inlet.

[0039] Optionally, the inlet is formed in the channel.

[0040] Optionally, the first and/or the second pair of seals include(s) one or more seals configured to withstand fluid pressures of 300 kPa and above. For example, the seals are configured to withstand fluid pressures of 300 kPa to 10,000 kPa, such as 300 kPa to 1,000 kPa, e.g. 300 kPa to 500 kPa.

[0041] In one embodiment, the first pair of seals includes a first seal configured to withstand fluid pressures of 300 kPa and above (such as 300 kPa to 10,000 kPa, such as 300 kPa to 1,000 kPa, e.g. 300 kPa to 500 kPa) and a second seal configured to withstand fluid pressures of up to 299 kPa (such as 50 kPa to 299 kPa).

[0042] In one embodiment, the second pair of seals includes first and second seals configured to withstand fluid pressures of up to 299 kPa (such as 50 kPa to 299 kPa).

[0043] Optionally, the second bearing comprises or consists solely of a ring fixed relative to an inner periphery of the rotary piston.

[0044] Optionally, the rotary piston internal combustion engine further comprises a lubricating substance reservoir and a pump. The pump may be configured to pump the lubricating substance from the lubricating substance reservoir to the first bearing.

[0045] Optionally, the rotary piston internal combustion engine further comprises a rotor cooling system cavity and a lubricating substance reservoir, wherein in use, the lubricating substance reservoir is pressurised by blow-by gases in the rotor cooling system cavity such that the lubricating substance flows to the first bearing from the lubricating substance reservoir.

[0046] Optionally, the lubricating substance reservoir is located internally or externally of the engine housing.

[0047] Optionally, the rotary piston has a substantially triangular shape with three apex seals. An inner peripheral surface of the engine housing may have a two lobed epitrochoidal shape. In use, the apex seals may maintain physical contact with the inner peripheral surface.

[0048] According to a second aspect of the invention, there is provided a vehicle or a device comprising a rotary piston internal combustion engine according to the first aspect of the invention.

[0049] The vehicle may be a car, a motorcycle, an aircraft (such as an unmanned aerial vehicle), jet skis or a snowmobile.

[0050] The device may be a powertool, such as a chainsaw.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0051] The present disclosure will now be described, by way of example only, with reference to the accompanying drawings, of which:

[0052] FIGS. 1a and 1b are cross-sectional views of an engine where a rotary piston is shown at different orientations with respect to the engine housing;

[0053] FIG. **2** is a perspective view of a rotary piston;
[0054] FIG. **3** is a cross-sectional view of a portion of an engine;
[0055] FIG. **4** is a perspective view of an output shaft;
[0056] FIG. **5** is a perspective view of a rotary piston mounted on an output shaft;
[0057] FIG. **6** is a perspective view of a bearing;
[0058] FIG. **7** is a cross-sectional view of a portion of an engine;
[0059] FIG. **8** is a cross-sectional view of a portion of an engine;
[0060] FIG. **9** is a cross-sectional view of an output shaft;
[0061] FIGS. **10a** and **10b** are cross-sectional views of first and second embodiments of the disclosure;
[0062] FIG. **11** is a cross-sectional view of part of an engine showing a bearing;
[0063] FIG. **12** is a schematic illustration of an engine according to a third embodiment of the disclosure; and
[0064] FIG. **13** is a schematic illustration of an engine according to a fourth embodiment of the disclosure.

DESCRIPTION OF THE INVENTION

[0065] Referring to FIGS. **1a** and **1b**, there is shown an engine **1** which includes an engine housing **10** configured to accommodate at least an output shaft **12** and a rotary piston **16**. The engine housing **10** defines a cavity having an inner peripheral surface **104**. The engine housing **10** may have an inlet port **100** and an exhaust port **102**. The inlet port **100** may be configured to supply air and/or fuel charge to the cavity. The exhaust port **102** may be configured to release exhaust gases, for example, to the atmosphere. The internal peripheral surface **104** may have a two lobed epitrochoidal shape. The rotary piston **16** may have a substantially triangular shape. However, it will be appreciated by the skilled person that alternative shapes of the internal peripheral surface **104** (e.g. a three or more lobed epitrochoidal shape) and the rotary piston **16** may be used without departing from the scope of the disclosure. The rotary piston **16** may have apex seals **160** at its apices. There may be one or more apex seals **160** located at each apex of the rotary piston **16**. During use, the apex seals **160** may be configured to maintain physical contact with lubricating substance on the internal peripheral surface **104**. In some versions, the apex seals **160** may be configured to maintain physical contact with the internal peripheral surface **104**. Combustion chambers **106** may be formed between the apex seals **160** and the internal peripheral surface **104**. During combustion, gases expand which increases the pressure in the combustion chambers **106**. The high-pressure gases move towards higher volume areas causing the rotary piston **16** to be driven relative to the engine housing **10**.

[0066] Referring to FIGS. **3** to **5**, the rotary piston **16** is mounted eccentrically on the output shaft **12**. As the rotary piston **16** is driven, the output shaft **12** also rotates relative to the engine housing **10** about an axis **14**. For every rotation by the rotary piston **16** the output shaft **12** rotates three times. Power can therefore be transmitted to parts to be operated, e.g. a wheel axle or a rotor blade.

[0067] A first bearing **18** is configured to support and facilitate rotation of the output shaft **12**. In use, the first bearing **18** may be located towards a first end **12a** of the output shaft **12**.

[0068] A second bearing **11** is configured to support and facilitate rotation of the rotary piston **16**. In some embodiments, the second bearing **11** may consist solely of a ring fixed relative to an inner periphery **162** of the rotary piston **16**.

[0069] The engine **1** may comprise a third bearing **15** which may be configured to support and facilitate rotation of the output shaft **12**. In use, the third bearing **15** may be located towards a second end **12b** of the output shaft **12**, opposite from the first end **12a**. The rotary piston **16** may be positioned along the axis **14** between the first bearing **18** and the third bearing **15**.

[0070] Referring to FIG. **6**, the first bearing **18** may include a ring **182** and a retainer **184**. The retainer **184** may be located within the ring **182** and mounted coaxially relative thereto. A rolling element **186** may be held relative to the ring **182** by the retainer **184**. The retainer **184** may define

an aperture **1840**. The aperture **1840** may extend from an outer surface of the retainer **184** to an inner surface of the retainer **184**. The aperture **1840** may be configured to receive at least a part of the rolling element **186**. In some embodiments, there may be a plurality of rolling elements **186** and respective apertures **1840**. During use, the rolling element(s) **186** may be in physical contact with the output shaft **12**. The ring **182** may include an inlet **1820** configured to receive a lubricating substance. The ring **182** may include a channel **1822** which extends around an outer surface thereof. The channel **1822** may be in fluid communication with the inlet **1820**. For example, the inlet **1820** may be formed in the channel **1822**.

[0071] In embodiments comprising a third bearing **15**, the third bearing **15** may have the same or a similar configuration as the first bearing **18**.

[0072] Referring back to FIG. 3, the first bearing **18** may be sandwiched between a first pair of seals **180**. The first pair of seals **180** may include one or more high pressure seals configured to withstand fluid pressures of 300 kPa and above. The one or more high pressure seals may be configured to withstand fluid pressures of 300 kPa to 10,000 kPa, such as of 300 kPa to 1,000 kPa, e.g. 300 kPa to 500 kPa. For example, the one or more high pressure seals may be multi (e.g. double) lip high fluid pressure seals. The first pair of seals may include one or more low pressure seals configured to withstand fluid pressures of up to 299 kPa. The one or more low pressure seals may be configured to withstand fluid pressures of 50 kPa to 299 kPa. For example, the one or more low pressure seals may be configured to withstand fluid pressures of 100 kPa to 299 kPa. In some embodiments of the disclosure, the first pair of seals **180** may include a high pressure seal **1800** configured to withstand fluid pressures of 300 kPa and above and a low pressure seal **1802** configured to withstand fluid pressures of up to 299 kPa (FIG. 7).

[0073] The second bearing **11** may be sandwiched between a second pair of seals **110**. The second pair of seals **110** may be low pressure seals configured to withstand fluid pressures of up to 299 kPa.

[0074] The third bearing **15** may be sandwiched between a pair of seals. The third bearing **15** may be sandwiched between a third seal **150** and a thrust bearing **17**. The third bearing **15** may be in fluid communication with the thrust bearing **17**. The third seal **150** may be a low pressure seal configured to withstand fluid pressures of up to 299 kPa.

[0075] The thrust bearing **17** may be a conventional rotary bearing, such as a ball bearing. The thrust bearing **17** may include a first side **170** and a second side **172** spaced apart from the first side **170**. The thrust bearing **17** may include an inlet **1700** and an outlet **1720** (FIG. 7).

[0076] Referring to FIG. 3, the output shaft **12** defines an internal path **13**. In use, the lubricating substance may flow from the first bearing **18** to the second bearing **11** along a first section **130** of the internal path **13**. The lubricating substance may be able to flow from the second bearing **11** to the third bearing **15** along a second section **132** of the internal path **13**.

[0077] The output shaft **12** may define a bore **120** which at least partially extends between the first end **12a** and the second end **12b** along the axis **14**.

[0078] In some embodiments, a rod **19** may be located within the bore **120**. The rod **19** may be hollow. The rod **19** may be comprised of aluminium. The rod **19** may have a first end **190** and a second end **192**.

[0079] The rod **19** may comprise a first barrier member **1200**. In use, the first barrier member **1200** may be located between the first section **130** and the second section **132**. The first barrier member **1200** prevents continued flow of lubricating substance along the bore **120** by redirecting flow from the first section **130** to the second bearing **11**. As illustrated, the first barrier member **1200** may comprise a thickening of the rod **19**. The first barrier member **1200** may have a conical shape. However, in some embodiments the first barrier member **1200** may take another form, such as a deflector part which extends from the rod **19**. In some embodiments, the first barrier member **1200** may not constitute a part of the rod **19**. For instance, the first barrier member **1200** may constitute another component of the engine **1**, such as an internal wall or part of the output shaft **12**.

[0080] The rod **19** may comprise a second barrier member **1202**. In use, the second barrier member **1202** may be located between the second section **132** and the second end **12b** of the output shaft **12**. The second barrier member **1202** prevents continued flow of lubricating substance along the bore **120** by redirecting flow from the second section **132** to the third bearing **15**. As illustrated, the second barrier member **1202** may comprise a thickening of the rod **19**. However, in some embodiments the second barrier member **1202** may take another form, such as a deflector part which extends from the rod **19**. In some embodiments, the second barrier member **1202** may not constitute a part of the rod **19**. For instance, the second barrier member **1202** may constitute another component of the engine **1**, such as an internal wall or part of the output shaft **12**.

[0081] The rod **19** serves two key purposes. First, the rod **19** comprises the first and second barrier members **1200**, **1202** which function as diverters to ensure that oil is redirected from the bore **120** to the respective second and third bearings **11**, **15**. Second, the presence of the rod **19** minimises the volume of oil required to prime the engine. This is because the rod **19** accommodates volume within the bore **120**. In versions where the rod **19** is hollow this serves to improve overall weight gains, which can be especially advantageous, e.g. in aerospace applications.

[0082] The rod **19** may be biased towards the first end **12a** of the output shaft **12**. The rod **19** may be biased resiliently by a spring **1206** acting on the second end **192** of the rod **19** (FIG. 7). The spring **1206** ensure that the rod **19** remains seated correctly during use. For example, the spring **1206** can account for differing thermal expansion of the output shaft **12** and the rod **19**.

[0083] The first section **130** and the second section **132** of the internal path **13** will now be described further with reference to FIG. 9. The first section **130** may include a first part **1300**, a second part **1302** and a third part **1304**. The second section **132** may include first part **1320**, a second part **1322** and a third part **1324**.

[0084] The first and second parts **1300**, **1302**; **1320**, **1322** of the respective first and second sections **130**; **132** may be substantially perpendicular relative to the axis **14**. The third parts **1304**; **1324** of the respective first and second sections **130**; **132** may be substantially parallel relative to the axis **14**.

[0085] Once a lubricating substance has been delivered to the first bearing **18** (described in more detail below), the lubricating substance can enter the first section **13** via the first part **1300** and flow along the third part **1304** until it is redirected into the second part **1302** by the first barrier member **1200**. The lubricating substance then flows into the second bearing **11** before being delivered, sequentially, to the first part **1320** and third part **1324** of the second section **132**, until it is redirected into the second part **1322** by the second barrier member **1202**. The lubricating substance then flows into the third bearing **15** and exits via the thrust bearing **17**.

[0086] In versions that do not include the rod **19**, the output shaft **12** may comprise two bores (not shown). A first bore may be drilled into the output shaft **12** from the first end **12a** and a second bore may be drilled into the output shaft **12** from the second end **12b**. In such versions, the first and second parts **1300**, **1302** of the first section **130** may be in fluid communication with the first bore. The first and second parts **1320**, **1322** of the second section **132** may also be in fluid communication with the second bore. The first barrier member **1200** may be defined by a section of the output shaft **12** that separates the first bore from the second bore.

[0087] Referring to FIGS. **10a** and **10b**, the engine **1** may further comprise a lubricating substance reservoir **2** and a pump **4**. The lubricating substance reservoir **2** may be located internally (FIG. **10a**) or externally (FIG. **10b**) of the engine housing **10**. Referring to FIG. **10b**, the lubricating substance reservoir **2** may have a calibrated sight glass **20** configured to allow a user to determine a lubricating substance level. The pump **4** may be configured to pump the lubricating substance from the lubricating substance reservoir **2** to the first bearing **18**. In some embodiments of the disclosure, the pump **4** may be controlled by an Engine Control Unit (ECU). The ECU may be configured to receive relevant information from sensors located in the engine **1**. The sensors may include temperature, pressure, level and/or flowrate sensors. Alternatively, the pump **4** may be manually

controlled.

[0088] Referring to FIG. 11, the lubricating substance may be supplied to the first bearing **18** via an inlet **108** and supply channel **1080**. The supply channel **1080** may supply the lubricating substance to the channel **1822** of the first bearing **18**.

[0089] Referring to FIG. 12, the engine **1** may further comprise a metering valve **6**. The metering valve **6** may be configured to control the amount of lubricating substance delivered to the internal peripheral surface **104** of the engine housing **10** to provide lubrication to the contacting surfaces of the rotary piston **16** itself.

[0090] Referring to FIG. 13, the engine **1** may further comprise a self-pressurising air rotor cooling system (SPARCS). The SPARCS may comprise a rotor cooling system cavity **81**.

[0091] Referring to FIG. 2, the rotary piston **16** may have a first face **168** and a second face **169**. The first face **168** and the second face **169** may be on opposing sides of the rotary piston **16**. One or more side seals **166** may be positioned on the first face **168** and/or the second face **169**. The side seal(s) **166** may be located between the apex seals **160**.

[0092] Referring to FIGS. 1a and 1b, the combustion gases in the working chambers **106** may leak past the side seal(s) **166** to passages located in the rotary piston **16**. These leaked combustion gases are referred to as blow-by gases. The blow-by gases may enter the rotor cooling system cavity **81** (FIG. 13) via the passages. Accordingly, the rotor cooling system cavity **81** may be pressurised. In such versions, the lubricating substance reservoir **2** may be pressurised by the blow-by gases in the rotor cooling system cavity **81** (indicated in FIG. 13 by a double headed arrow designated A).

Accordingly, pure oil from the lubricating substance reservoir **2** may be forced to the first bearing **18** (indicated in FIG. 13 by arrow designated B) by the pressure acting on the lubricating substance reservoir **2**. Thus, the pump **4** (shown on FIG. 12) may not be required.

[0093] A separate feed directly from the lubricating substance reservoir **2** to the internal peripheral surface **104** of the engine may be provided, although this is unlikely to be necessary because initial testing has shown that sufficient lubricating substance can reach all of the essential components of the engine via the internal path of the output shaft **12**.

[0094] Referring to FIGS. 10a and 10b, the pump **4** pumps the lubricating substance from the lubricating substance reservoir **2** to the internal path **13** of the engine housing **10** via the first bearing **18**. In one embodiment of the disclosure, the pump **4** is controlled by the Engine Control Unit (ECU). The sensors located in the engine **1** send relevant information (for example, oil level, pressure, temperature and flowrate) regarding the physical state of the engine **1** to the ECU. The relevant information may be sent at pre-determined intervals. The flowrate of the lubricating substance supplied to the first bearing **18** is then adjusted by the ECU based on the relevant information to compensate for any variance in the physical properties of the lubricating substance, for example, viscosity. For example, at higher temperatures, the viscosity of the lubricating substance inside the engine **1** is likely to decrease, therefore, lower flowrates are required to achieve sufficient lubrication. The ECU ensures optimum amount of lubricating substance is supplied to the engine, thus limiting wastage of the lubricating substance. In another embodiment of the disclosure, the pump **4** is manually controlled. The user can determine the amount of lubricating substance consumed in a time period by observing change in the lubricating substance level from the calibrated sight glass **20** on the lubricating substance reservoir **2**. Alternatively, the sensors installed in the engine **1** provide the relevant information (for example, remaining volume of lubricating substance, pressure, temperature and flowrate) to the user via a user interface. Subsequently, the user can alter the flowrate of the lubricating substance accordingly by adjusting power of the pump **4**.

[0095] In some embodiments, referring particularly to FIGS. 6 and 11, the lubricating substance is pumped to the inlet **108** of the engine housing **10**. The lubricating substance then flows along the supply channel **1080** until it reaches the channel **1822** of the first bearing **18**. The lubricating substance flows along the channel **1822** until it enters the inlet **1820**. The lubricating substance then

lubricates contact surfaces between the rolling elements **186** and the output shaft **12**. As the output shaft **12** rotates, the rolling elements **186** also rotate in the same direction. The lubricating substance is further distributed between the contact surfaces. The axial vibration of the output shaft **12** is thus reduced by the first bearing **18**, ensuring smooth rotation along the axis **14**.

[0096] The first pair of seals **180** are provided to direct flow of the lubricating substance from the first bearing **18** to the first section **130** of the internal path **13** via the first part **1300**. Similarly, the second pair of seals **110** are provided to direct flow of the lubricating substance from the second bearing **11** to the second section **132** of the internal path **13** via the first part **1320**. The third seal **150** is also provided to direct flow the lubricating substance in the third bearing **15** to the inlet on the first side **1700** of the thrust bearing **17**. In some embodiments, the third bearing **15** and the thrust bearing **17** are configured so that the lubricating substance is received by the thrust bearing **17** along a direction parallel to the axis **14**. The lubricating substance then lubricates the thrust bearing **17** and exits from the outlet of the second side **1720** of the thrust bearing **17**.

[0097] The following advantages of the engine **1** are apparent.

[0098] First, lubricating substance consumption by the engine **1** is significantly reduced when compared to the prior art. This is because lubricating substance is directed specifically to locations that require lubrication, i.e., the first **18**, second **11** and third **15** bearings, and not to component parts that require little or no lubrication. This controlled feed of the lubricating substance into the engine **1** ensures minimum amount of wastage. Engines according to the present invention have been shown to consume oil at a rate of approximately 10 cc/hr, compared to approximately 150 cc/hr in the prior art. Accordingly, engines of the present disclosure have lower hydrocarbon emissions when compared to the prior art, and are therefore more environmentally friendly than prior art engines.

[0099] Second, the bearings receive pure oil or substantially pure oil due to the lubricating substance being delivered specifically to those locations first. The lubricating substance does not mix with impurities inside the engine **1**, e.g. from the sliding surfaces of the rotary piston, before contacting the bearings. This can lead to an increase in the lifetime of these moving components.

[0100] Third, since a significantly smaller volume of lubricating substance is required, the overall size and/or weight of the vehicle or device can be reduced, e.g. by reducing the volume of the lubricating substance reservoir. This advantage is particularly beneficial where the engine is utilised in aircraft, such as unmanned aerial vehicles.

[0101] Fourth, having a controlled flow path for the lubricating substance simplifies the control measures that are required—there is only one inlet and outlet for the lubricating substance. These simplified control measures can result in more accurate controls being employed by the ECU, further increasing the reliability of the engine **1**.

[0102] It will be readily appreciated by the skilled person that additional bearings may be provided that are configured to support and facilitate rotation of the output shaft **12**. Furthermore, the engine **1** may comprise one or more additional rotary pistons **16**. Such rotary pistons may comprise associated respective bearings, as would be appreciated by the skilled person.

[0103] Furthermore, whilst embodiments depicted in the figures show that the lubricating substance is delivered to the first bearing **18** from the lubricating substance reservoir **2**, it will be appreciated by the skilled person that the lubricating substance could alternatively be delivered to any one of the other bearings, e.g. the second bearing **11** or the third bearing **15**, without departing from the scope of the invention.

[0104] When used in this specification and claims, the terms “comprises” and “comprising” and variations thereof mean that the specified features, steps or integers are included. The terms are not to be interpreted to exclude the presence of other features, steps or components.

[0105] The invention may also broadly consist in the parts, elements, steps, examples and/or features referred to or indicated in the specification individually or collectively in any and all combinations of two or more said parts, elements, steps, examples and/or features. In particular,

one or more features in any of the embodiments described herein may be combined with one or more features from any other embodiment(s) described herein.

[0106] Protection may be sought for any features disclosed in any one or more published documents referenced herein in combination with the present disclosure.

[0107] Although certain example embodiments of the invention have been described, the scope of the appended claims is not intended to be limited solely to these embodiments. The claims are to be construed literally, purposively, and/or to encompass equivalents.

Claims

1. A rotary piston internal combustion engine including: an engine housing; an output shaft configured to rotate relative to the engine housing about an axis for transmitting power to parts to be operated; a rotary piston mounted eccentrically on the output shaft; a first bearing configured to support and facilitate rotation of the output shaft, the first bearing being sandwiched between a first pair of seals; and a second bearing configured to support and facilitate rotation of the rotary piston, the second bearing being sandwiched between a second pair of seals; wherein the output shaft defines an internal path comprising a first section to allow a lubricating substance to flow between the first bearing and the second bearing, in use.
2. A rotary piston internal combustion engine according to claim 1, wherein the internal path comprises a second section to allow the lubricating substance to flow between the second bearing and a third bearing, in use.
3. A rotary piston internal combustion engine according to claim 2, wherein the output shaft defines a bore which at least partially extends along the axis, the rotary piston internal combustion engine further comprising a first barrier member, the first barrier member being located in the bore between the first section and the second section, wherein the first barrier member is configured to direct flow of the lubricating substance from the first section into the second bearing.
4. A rotary piston internal combustion engine according to claim 3, wherein the first barrier member is configured to prevent flow of the lubricating substance along the bore and past the first barrier member.
5. A rotary piston internal combustion engine according to claim 3, further comprising a second barrier member, wherein the second barrier member is located in the bore and is configured to direct flow of the lubricating substance from the second section into the third bearing, preferably wherein the second barrier member is configured to prevent flow of the lubricating substance along the bore and past the second barrier member.
6. (canceled)
7. A rotary piston internal combustion engine according to claim 3, further comprising a rod located within the bore, preferably wherein the rod comprises one or both of the first barrier member and a second barrier member.
8. (canceled)
9. A rotary piston internal combustion engine according to claim 5, wherein the rod is biased towards the first bearing.
10. A rotary piston internal combustion engine according to claim 2, wherein the third bearing is configured to support and facilitate rotation of the output shaft.
11. A rotary piston internal combustion engine according to claim 2, wherein the third bearing is in fluid communication with a thrust bearing.
12. A rotary piston internal combustion engine according to claim 11, wherein the second bearing and the thrust bearing are configured so that, in use, the lubricating substance is received by the thrust bearing from the second bearing along a direction parallel to the axis.
13. A rotary piston internal combustion engine according to claim 11, the thrust bearing including a first side and a second side, the second side being spaced apart from the first side along the axis,

- wherein the first side includes an inlet for receiving the lubricating substance from the third bearing and the second side includes an outlet for the lubricating substance to exit the thrust bearing.
- 14.** A rotary piston internal combustion engine according to claim 11, wherein the third bearing is sandwiched between the thrust bearing and a third seal.
- 15.** A rotary piston internal combustion engine according to claim 1, wherein the first bearing and/or the third bearing, where present, includes: a ring; a retainer located within the ring and mounted coaxially relative thereto; and a rolling element, wherein the rolling element is held relative to the ring by the retainer.
- 16.** A rotary piston internal combustion engine according to claim 15, wherein the retainer defines an aperture which extends from an outer surface to an inner surface thereof and wherein the aperture receives at least a part of the rolling element.
- 17.** A rotary piston internal combustion engine according to claim 15, wherein the ring includes an inlet configured to receive the lubricating substance, preferably wherein the ring includes a channel which extends around an outer surface thereof, the channel being in fluid communication with the inlet, more preferably wherein the inlet is formed in the channel.
- 18.** (canceled)
- 19.** (canceled)
- 20.** A rotary piston internal combustion engine according to claim 1, wherein the first and/or the second pair of seals includes one or more seals configured to withstand fluid pressures of 300 kPa and above, preferably wherein the first pair of seals includes a first seal configured to withstand fluid pressures of 300 kPa and above and a second seal configured to withstand fluid pressures of up to 299 kPa.
- 21.** (canceled)
- 22.** A rotary piston internal combustion engine according to claim 1, wherein the second bearing comprises a ring fixed relative to an inner periphery of the rotary piston.
- 23.** A rotary piston internal combustion engine according to claim 1 further comprising a lubricating substance reservoir and a pump, wherein the pump is configured to pump the lubricating substance from the lubricating substance reservoir to the first bearing.
- 24.** A rotary piston internal combustion engine according to claim 1, further comprising a rotor cooling system cavity and a lubricating substance reservoir, wherein in use, the lubricating substance reservoir is pressurised by blow-by gases in the rotor cooling system cavity such that the lubricating substance flows to the first bearing from the lubricating substance reservoir.
- 25.** A vehicle or a device comprising a rotary piston internal combustion engine according to claim 1.
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